Intro to ML - VisionGATE Solutions

GATE Data Science and Artificial Intelligence (GATE DA)

Topics Index

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Document compiled for GATE DA Preparation

1 Task Taxonomy

Key idea for this section

Every question here is answered by asking two things in order:

1. **Does the system learn from data?** If a human wrote the rules and the system never improves with experience, it is not machine learning. This is Mitchell's definition used as a test: a program learns from experience E with respect to a task T and performance measure P if its performance at T, as measured by P, improves with E.
2. **Is the target continuous or discrete?** Continuous target means regression. Discrete target means classification. No labels at all means unsupervised.

Q1. In the context of machine learning, what is a *model*?
Answer: (C)

Solution. A model is a mathematical representation of reality — a function h_θ mapping inputs to outputs, together with parameters θ learned from data. For example, $\hat{y} = \mathbf{w}^\top \mathbf{x} + \mathbf{b}$ is a linear model.

Option (A) confuses the model with the data used to fit it. Option (B) is wrong because a model is an abstract mathematical object, not hardware; the same model can run on any machine. Option (D) confuses the model with the target values it is trained against.

Q2. Which of the following is a regression problem?
Answer: (D)

Solution. House price is a *continuous* quantity — it can take any real value in a range, and values like 68.4 lakh are meaningful. A continuous target means regression.

Options (A), (B) and (C) all have discrete targets: spam or not spam, cat or dog, churn or no churn. These are classification problems.

Q3. Which of the following is an example of a machine learning problem?
Answer: (D)

Solution. Apply Mitchell's test. For spam detection we can name all three components: T is classifying emails as spam or not spam, P is the fraction classified correctly, and E is a corpus of emails already labelled by users. The system genuinely improves as more labelled emails arrive.

The other three are fixed algorithms. Computing an average, sorting records, and Dijkstra's shortest path all produce provably correct output every time, and their performance on the millionth run is identical to the

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first. There is no E, so there is no learning. Note that Dijkstra's algorithm *is* artificial intelligence in the broad sense — search was a founding problem of the field — but it is not machine learning.

Q4. What kind of values do the parameters of a regression model take?

Answer: (C)

Solution. The parameters of a regression model are the weights $\mathbf{w}$ and bias b in $\hat{y} = \mathbf{w}^\top \mathbf{x} + b$, and these are real valued. This is essential: gradient-based optimisation requires parameters to live in a continuous space so that we can take derivatives and make arbitrarily small updates.

Option (A) is wrong because binary weights would make the model a discrete combinatorial object with no gradients. Option (B) is wrong because negative weights are essential — they encode features that *decrease* the prediction. Option (D) confuses parameters with probabilities; a probability is constrained to $[0, 1]$, but a weight is not.

Q5. Which of the following are **classification** problems? (Select all that apply.)

Answer: (A, C)

Solution. In both (A) and (C) the target variable is discrete. Option (A) is binary classification with two classes, benign and malignant. Option (C) is multi-class classification with five categories.

Options (B) and (D) have continuous targets — rainfall in millimetres and resale value in currency — so they are regression problems.

Note. *A useful test: ask whether the average of two valid target values is itself a valid target value. The average of 12mm and 18mm of rainfall is 15mm, which is meaningful, so the target is continuous. The average of "benign" and "malignant" is meaningless, so the target is discrete.*

Q6. Which of the following is a regression problem?

Answer: (A)

Solution. Temperature in degrees Celsius is continuous, so predicting it is regression.

Options (B), (C) and (D) are all classification: fraudulent or not (binary), which of several languages (multi-class), and pass or fail (binary).

Q7. Which of the following are **unsupervised** learning problems? (Select all that apply.)

Answer: (A, B)

Solution. Unsupervised learning operates on inputs $\{\mathbf{x}^{(i)}\}$ with *no* accompanying labels, and the goal is to discover structure rather than to predict a known target.

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Option (A) is clustering. Nobody has told the algorithm what the correct segments are; it must discover groupings from the purchase data itself. Option (B) is dimensionality reduction, where the goal is a compact representation, not a prediction against a label.

Options (C) and (D) both state explicitly that labelled data is available — past sale prices and labelled digit images — so both are supervised.

Note. *Be careful with option (B). Dimensionality reduction does have a reconstruction objective, so it might look supervised. But the "target" is the input itself, not an external label supplied by a human, so it is unsupervised. When targets are generated automatically from the data's own structure, the modern term is* self-supervised *learning; this is how large language models are pretrained.*

Q8. A model outputs $\hat{y} \in \{-1, +1\}$ for each input. What kind of problem is this?

Answer: (D)

Solution. The output takes exactly two distinct values, so this is binary classification. The particular choice of $\{-1, +1\}$ rather than $\{0, 1\}$ is a labelling convention used by algorithms such as the support vector machine and the perceptron, because it makes the margin condition $y\,(\mathbf{w}^\top \mathbf{x} + b) > 0$ compact to write. The convention does not change the nature of the problem.

Option (A) is wrong because the output is not continuous. Options (B) and (C) are unsupervised methods, and this model is clearly producing a prediction for each input.

Q9. A model predicts a student's grade as one of A, B, C, D, or F. What kind of problem is this?

Answer: (B)

Solution. The output has five distinct levels, so the target is discrete and this is multi-class classification.

The tempting wrong answer is (A), because grades are *ordered*: A is better than B, which is better than C. But ordering alone does not make a variable continuous. There is no grade strictly between A and B, and the "distance" from A to B is not a well-defined number.

Note. *Problems with an ordered discrete target have their own name —* ordinal regression *— and specialised methods exist that exploit the ordering. For an introductory course, treating this as classification is correct.*

Q10. Problem 1: predict whether a loan applicant will default (yes/no). Problem 2: predict the amount of the loan a bank should sanction, in rupees.

What is the key difference?

Answer: (C)

Solution. Problem 1 has a discrete target with two values, so it is classification. Problem 2 has a continuous target measured in rupees, so it is regression. This is the essential difference.

Option (A) is wrong because both problems are supervised — both require historical labelled records. Option (B) is wrong because Problem 1 is not regression at all. Option (D) is wrong because both problems require labelled data; you cannot learn to predict sanctioned amounts without records of past sanctioned amounts.

2 Norms

Key ideas for this section

A norm is a function $f : \mathbb{R}^d \to \mathbb{R}$ satisfying, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ and all $\alpha \in \mathbb{R}$:

	Axiom	Name
(a)	$f(\mathbf{x}) \geq 0$	non-negativity
(b)	$f(\mathbf{x}) = 0 \iff \mathbf{x} = \mathbf{0}$	definiteness
(c)	$f(\mathbf{x}+\mathbf{y}) \leq f(\mathbf{x}) + f(\mathbf{y})$	triangle inequality
(d)	$f(\alpha\mathbf{x}) = \lvert\alpha\rvert\, f(\mathbf{x})$	absolute homogeneity

The three norms used throughout:

$$\|\mathbf{x}\|_1 = \sum_{i=1}^{d} |x_i|, \qquad \|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^{d} x_i^2}, \qquad \|\mathbf{x}\|_\infty = \max_i |x_i|$$

Two facts used repeatedly below:

- **Monotonicity in p.** For $p \leq q$ we have $\|\mathbf{x}\|_p \geq \|\mathbf{x}\|_q$. In particular $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$.
- **Ratio bounds.** $1 \leq \dfrac{\|\mathbf{x}\|_1}{\|\mathbf{x}\|_2} \leq \sqrt{d}$, with the lower bound attained when exactly one entry is non-zero and the upper bound when all entries have equal magnitude.

Q1. If $\|\mathbf{v}\| = 5$ and $\|\mathbf{w}\| = 3$, what are the smallest and largest possible values of $\|\mathbf{v} - \mathbf{w}\|$ and of $\mathbf{v} \cdot \mathbf{w}$?

Answer: (B)

Solution. Upper bound on $\|\mathbf{v} - \mathbf{w}\|$. By the triangle inequality and homogeneity,

$$\|\mathbf{v} - \mathbf{w}\| \leq \|\mathbf{v}\| + \|-\mathbf{w}\| = \|\mathbf{v}\| + |-1|\,\|\mathbf{w}\| = 5 + 3 = 8$$

This is attained when $\mathbf{v}$ and $\mathbf{w}$ point in *opposite* directions.

Lower bound on $\|\mathbf{v} - \mathbf{w}\|$. By the reverse triangle inequality, $\|\mathbf{a}+\mathbf{b}\| \geq \big|\,\|\mathbf{a}\| - \|\mathbf{b}\|\,\big|$, so

$$\|\mathbf{v} - \mathbf{w}\| \geq \big|\,\|\mathbf{v}\| - \|\mathbf{w}\|\,\big| = |5 - 3| = 2$$

This is attained when $\mathbf{v}$ and $\mathbf{w}$ point in the *same* direction. The lower bound is not 0: two vectors of unequal length can never cancel completely.

Bounds on the dot product. Using $\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \, \|\mathbf{w}\| \cos \alpha$ and $\cos \alpha \in [-1, 1]$,

$$-15 \le \mathbf{v} \cdot \mathbf{w} \le 15$$

The maximum $+15$ occurs at $\alpha = 0$ (same direction) and the minimum -15 at $\alpha = \pi$ (opposite directions).

Option (A) wrongly takes the lower bound as 0 and forces the dot product to be non-negative. Option (C) gets the norm range right but forgets that $\cos \alpha$ can be negative. Option (D) confuses the range of the difference with the individual norms.

Q2. Let $\mathbf{x} = [3, \, -4, \, 12, \, 0]^\top$. What are $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$ and $\|\mathbf{x}\|_\infty$ respectively?

Answer: (A)

Solution.

$$\|\mathbf{x}\|_1 = |3| + |-4| + |12| + |0| = 3 + 4 + 12 + 0 = 19$$

$$\|\mathbf{x}\|_2 = \sqrt{3^2 + (-4)^2 + 12^2 + 0^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$$

$$\|\mathbf{x}\|_\infty = \max\{3, \, 4, \, 12, \, 0\} = 12$$

Sanity check against monotonicity: $12 \le 13 \le 19$. ✓

Option (B) forgets to take the square root, reporting $\|\mathbf{x}\|_2^2 = 169$. Option (C) sums the signed values $3 - 4 + 12 + 0 = 11$ instead of the absolute values. Option (D) takes the first entry rather than the largest.

Q3. Let $\mathbf{x} = [-1, \, 2, \, -2, \, 4, \, -4]^\top$. Which of the following is the **greatest**?

Answer: (A)

Solution. Computing all four:

$$\|\mathbf{x}\|_1 = 1 + 2 + 2 + 4 + 4 = 13$$

$$\|\mathbf{x}\|_2 = \sqrt{1 + 4 + 4 + 16 + 16} = \sqrt{41} \approx 6.4031$$

$$\|\mathbf{x}\|_3 = (1 + 8 + 8 + 64 + 64)^{1/3} = 145^{1/3} \approx 5.2536$$

$$\|\mathbf{x}\|_\infty = \max\{1, 2, 2, 4, 4\} = 4$$

So L_1 is greatest.

No computation is actually needed. The p-norm is *non-increasing* in p, so the smallest p always gives the largest value, and $p = 1$ is the smallest available here.

Note. *Why the norm decreases as p grows: normalise so that $\|\mathbf{u}\|_p = 1$. Then every component satisfies $|u_i| \le 1$, and raising a number in $[0, 1]$ to a* higher power *makes it* smaller. *Summing, $\|\mathbf{u}\|_q^q \le \|\mathbf{u}\|_p^p = 1$ for $q > p$. Intuitively, higher powers emphasise the largest component and suppress the rest, until at $p = \infty$ only the maximum survives.*

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Q4. For $\mathbf{x} = [1,\ 1,\ 1,\ 1]^\top$, what is the value of the ratio $\|\mathbf{x}\|_1/\|\mathbf{x}\|_2$?
Answer: (B)

Solution.

$$\|\mathbf{x}\|_1 = 1+1+1+1 = 4, \qquad \|\mathbf{x}\|_2 = \sqrt{1+1+1+1} = \sqrt{4} = 2$$
$$\frac{\|\mathbf{x}\|_1}{\|\mathbf{x}\|_2} = \frac{4}{2} = 2$$

Note that $2 = \sqrt{4} = \sqrt{d}$, so this vector attains the *maximum* possible ratio in $\mathbb{R}^4$. That is exactly what the theory predicts: the upper bound $\sqrt{d}$ is achieved when all components have equal magnitude.

Option (C) reports $\|\mathbf{x}\|_1$ alone. Option (D) inverts the ratio.

Q5. For which of the following vectors in $\mathbb{R}^4$ is the ratio $\|\mathbf{x}\|_1/\|\mathbf{x}\|_2$ the **smallest**?
Answer: (B)

Solution. Computing all four:

	$\mathbf{x}$	$\|\mathbf{x}\|_1$	$\|\mathbf{x}\|_2$	Ratio
(A)	$[1,1,1,1]^\top$	4	2	2.0000
(B)	$[3,0,0,0]^\top$	3	3	**1.0000**
(C)	$[2,2,1,0]^\top$	5	3	1.6667
(D)	$[1,1,1,0]^\top$	3	$\sqrt{3}$	1.7321

Option (B) has exactly one non-zero entry, so the two norms coincide and the ratio is exactly 1 — the theoretical minimum.

The reason: expanding the square of the L_1 norm,

$$\|\mathbf{x}\|_1^2 = \Big(\sum_i |x_i|\Big)^2 = \underbrace{\sum_i |x_i|^2}_{\|\mathbf{x}\|_2^2} + \underbrace{2\sum_{i<j} |x_i|\,|x_j|}_{\geq 0}$$

The cross terms are non-negative, so $\|\mathbf{x}\|_1 \geq \|\mathbf{x}\|_2$ always. Equality requires every cross term to vanish, which means no two components are simultaneously non-zero.

Note. *This ratio is a* sparsity measure*: it equals 1 for maximally concentrated vectors and $\sqrt{d}$ for maximally spread ones. This is the algebraic counterpart of the geometric fact that L_1 regularisation produces sparse solutions.*

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Q6. Let $\mathbf{u} = [2,\ -3,\ 6]^\top$ and $\mathbf{v} = [5,\ 1,\ 2]^\top$. What are the L_1, L_2 and L_∞ distances between $\mathbf{u}$ and $\mathbf{v}$?

Answer: (A)

Solution. Any norm induces a distance via $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$. First compute the difference:

$$\mathbf{u} - \mathbf{v} = [2-5,\ -3-1,\ 6-2]^\top = [-3,\ -4,\ 4]^\top$$

Then:

$$\|\mathbf{u} - \mathbf{v}\|_1 = 3 + 4 + 4 = 11$$

$$\|\mathbf{u} - \mathbf{v}\|_2 = \sqrt{9 + 16 + 16} = \sqrt{41} \approx 6.4031$$

$$\|\mathbf{u} - \mathbf{v}\|_\infty = \max\{3,\ 4,\ 4\} = 4$$

Check: $4 \leq 6.4031 \leq 11$. ✓

Option (B) forgets the square root. Option (C) adds the vectors instead of subtracting. Option (D) swaps the L_1 and L_∞ values.

Note. *The direction of subtraction does not matter for distance. We have $\mathbf{v} - \mathbf{u} = [3, 4, -4]^\top$, which points the opposite way, but all three norms are unchanged because $\|-\mathbf{z}\| = |-1|\,\|\mathbf{z}\| = \|\mathbf{z}\|$ by homogeneity. This is exactly the symmetry property $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$ that any distance must satisfy.*

Q7. Which of the following is true for **every** non-zero vector $\mathbf{x} \in \mathbb{R}^d$?

Answer: (B)

Solution. The p-norm is non-increasing in p, so larger p gives a smaller (or equal) value:

$$\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$$

Verify with $\mathbf{x} = [3, -4, 12, 0]^\top$ from Q2: $12 \leq 13 \leq 19$. ✓

Option (A) reverses the ordering. Option (C) misplaces $\|\mathbf{x}\|_\infty$. Option (D) is wrong because the ordering is universal, not data-dependent — it holds for every vector in every dimension.

Note. *Equality throughout occurs precisely when at most one component is non-zero. For $\mathbf{x} = [0, 0, 7]^\top$, all three norms equal 7.*

Q8. For a non-zero vector $x \in \mathbb{R}^d$, what is the range of the ratio $\|x\|_1/\|x\|_2$?

Answer: (B)

Solution. Lower bound. From the expansion in Q5, $\|\mathbf{x}\|_1^2 = \|\mathbf{x}\|_2^2 + 2\sum_{i<j} |x_i||x_j| \geq \|\mathbf{x}\|_2^2$, so the ratio is at least 1. Attained when exactly one entry is non-zero.

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Upper bound. Write the L_1 norm as a dot product with the all-ones vector $\mathbf{1} = [1, 1, \dots, 1]^\top$:

$$\|\mathbf{x}\|_1 = \sum_i |x_i| = \mathbf{1}^\top |\mathbf{x}|$$

By Cauchy–Schwarz,

$$\mathbf{1}^\top |\mathbf{x}| \;\le\; \|\mathbf{1}\|_2 \, \big\||\mathbf{x}|\big\|_2 \;=\; \sqrt{d}\, \|\mathbf{x}\|_2$$

since $\|\mathbf{1}\|_2 = \sqrt{1 + 1 + \cdots + 1} = \sqrt{d}$. Cauchy–Schwarz is tight when the two vectors are parallel, i.e. $|\mathbf{x}| = c\,\mathbf{1}$, meaning all components have equal magnitude.

Therefore $1 \le \|\mathbf{x}\|_1 / \|\mathbf{x}\|_2 \le \sqrt{d}$.

Option (A) overstates the upper bound. Option (C) misplaces the lower bound. Option (D) is impossible, since the ratio can never fall below 1.

Note. *An alternative derivation avoids Cauchy–Schwarz entirely. Let m and v be the mean and variance of the values $|x_i|$. Then $\|\mathbf{x}\|_1^2 = d^2 m^2$ and $\|\mathbf{x}\|_2^2 = d(m^2 + v)$, so*

$$\frac{\|\mathbf{x}\|_1^2}{\|\mathbf{x}\|_2^2} = d \cdot \frac{m^2}{m^2 + v} \le d$$

since $v \ge 0$. Equality holds exactly when $v = 0$, i.e. all the $|x_i|$ are equal. The variance of the components *is precisely what drags the ratio below $\sqrt{d}$.*

Q9. Define $f(x)$ = the number of non-zero entries of x (the "L_0 norm"). Exactly one norm axiom fails. Which one?

Answer: (A)

Solution. Absolute homogeneity fails. Take $\mathbf{x} = [1,\, 0]^\top$ and $\alpha = 2$. Then $2\mathbf{x} = [2,\, 0]^\top$, which still has exactly one non-zero entry, so

$$f(2\mathbf{x}) = 1 \qquad \text{but} \qquad |2|\, f(\mathbf{x}) = 2 \times 1 = 2$$

These differ, so $f(\alpha\mathbf{x}) \ne |\alpha| f(\mathbf{x})$. The reason is structural: scaling a vector by a non-zero constant never changes *which* entries are non-zero, so f is completely blind to magnitude.

The other axioms all hold, which is why option (D) is wrong:

– **Definiteness holds.** $f(\mathbf{x}) = 0$ means zero entries are non-zero, i.e. every entry is zero, i.e. $\mathbf{x} = \mathbf{0}$. The converse is immediate. So option (C) is false — there are no non-zero vectors with $f(\mathbf{x}) = 0$.

– **Triangle inequality holds.** If $x_i + y_i \neq 0$ then at least one of x_i, y_i is non-zero. Hence the support of $\mathbf{x} + \mathbf{y}$ is contained in the union of the supports, giving $f(\mathbf{x} + \mathbf{y}) \leq f(\mathbf{x}) + f(\mathbf{y})$.

Note. *This single failure has real consequences. Because $\|\cdot\|_0$ is not a norm, its "unit ball" is a union of coordinate axes rather than a convex set, and L_0-regularised regression is combinatorially hard — one must search over all subsets of features. L_1 regularisation is used instead precisely because it is convex while still promoting sparsity.*

Q10. Which of the following functions on $\mathbb{R}^d$ are valid norms? (Choose all the correct answers.)

Answer: (A, D)

Solution. (A) $f(\mathbf{x}) = \max_i |x_i|$ — **valid.** This is the L_∞ norm. Definiteness: the max of non-negative numbers is zero iff all are zero. Homogeneity: $\max_i |\alpha x_i| = |\alpha| \max_i |x_i|$. Triangle inequality: letting k be the index achieving the maximum on the left,

$$\|\mathbf{x} + \mathbf{y}\|_\infty = |x_k + y_k| \leq |x_k| + |y_k| \leq \max_i |x_i| + \max_i |y_i|$$

(B) Number of non-zero entries — not valid. Fails absolute homogeneity, as shown in Q9.

(C) $f(\mathbf{x}) = \left(\sum_i |x_i|^{1/2}\right)^2$ — **not valid.** This is the $p = 1/2$ quasi-norm, and it fails the triangle inequality. Take $\mathbf{x} = [1,\ 0]^\top$ and $\mathbf{y} = [0,\ 1]^\top$:

$$f(\mathbf{x} + \mathbf{y}) = f([1, 1]^\top) = (1+1)^2 = 4$$
$$f(\mathbf{x}) + f(\mathbf{y}) = 1 + 1 = 2$$

Since $4 > 2$, the triangle inequality is violated.

(D) $f(\mathbf{x}) = \sum_i |x_i|$ — **valid.** This is the L_1 norm. All three axioms follow componentwise from the corresponding properties of the scalar absolute value; in particular the triangle inequality follows from $|a + b| \leq |a| + |b|$ applied term by term.

Note. *The unit ball picture summarises this neatly. The L_1 diamond and the L_∞ square are both convex, as the triangle inequality requires. The $p = 1/2$ "ball" is a concave star shape, and the L_0 "ball" is a union of axes — neither is convex, and neither comes from a norm.*

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3 Vectorization and Dimensions

Key ideas for this section

- A colour (RGB) image of size $H \times W$ flattens to a vector of length $H \times W \times 3$. The factor of 3 is the three colour channels; a grayscale image has only one channel, so its length is $H \times W$.
- In $\mathbf{y} = A\mathbf{x} + \mathbf{b}$ with $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ and $\mathbf{y} \in \mathbb{R}^{d_{\text{out}}}$, the matrix A has shape $d_{\text{out}} \times d_{\text{in}}$ and the bias $\mathbf{b} \in \mathbb{R}^{d_{\text{out}}}$. **Shape is always (output $\times$ input).**
- The design matrix X has shape $n \times d$: **rows are samples, columns are features.** Note this is the transpose of the convention used for A, which is why the two are so easily confused.

Q1. A colour (RGB) image of size 512×512 is fed to an algorithm that classifies it into one of 10 categories.

(i) If the input image is flattened into a vector $\mathbf{x}$, what is its length?

Answer: (B)

Solution.

$$512 \times 512 \times 3 = 262144 \times 3 = 786432$$

Option (A) is $512 \times 512 = 262144$, forgetting the three colour channels — this is the length for a *grayscale* image of the same size. Option (C) is 512×3, treating the image as one-dimensional. Option (D) is a single side length.

(ii) If the output is a vector $\mathbf{y}$ of class scores, what is its length?

Answer: (C)

Solution. One score per class, and there are 10 categories, so $\mathbf{y} \in \mathbb{R}^{10}$.

Option (A) would be a single scalar, which is what a *regression* model outputs, not a 10-class classifier. Option (B) confuses the number of classes with the number of colour channels. Option (D) multiplies the two, which has no meaning here.

(iii) The model is $\mathbf{y} = A\mathbf{x} + \mathbf{b}$. How many elements does the matrix A contain?

Answer: (C)

Solution. A maps $\mathbb{R}^{786432} \to \mathbb{R}^{10}$, so its shape is 10×786432 (output $\times$ input). The number of elements is

$$10 \times 786432 = 7864320$$

Option (A) adds the two dimensions instead of multiplying. Option (D) is the answer to part (iv) — it includes the 10 bias terms, which are not part of A.

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(iv) How many learnable parameters does the model have in total, counting both A and $\mathbf{b}$?

Answer: (B)

Solution. The matrix contributes 7864320 parameters and the bias vector contributes one parameter per output, so $\mathbf{b} \in \mathbb{R}^{10}$ adds 10 more:

$$7864320 + 10 = 7864330$$

Option (A) omits the bias entirely. Option (C) adds only 2, as though there were a single scalar bias.

Note. *This is why fully connected layers are avoided for raw images. A single linear layer here needs nearly 8 million parameters, which is enormous for one layer. Convolutional layers exploit spatial structure and weight sharing to achieve the same task with a few thousand parameters, which was one of the key insights behind LeNet and AlexNet.*

(v) The model is trained on a batch of 500 such images. If the design matrix X stores one image per row, how many entries does X contain in total?

Answer: (A)

Solution. With one image per row, X has shape 500×786432, so

$$500 \times 786432 = 393216000$$

Option (B) is 2×786432. Option (C) adds rather than multiplies. Option (D) uses the grayscale length 262144 instead of the RGB length.

Q2. A video clip consists of 60 frames, each a colour (RGB) image of size 128×128.

(i) Flattening the entire clip into a single vector, what is its length?

Answer: (B)

Solution. Multiply frames by height by width by channels:

$$60 \times 128 \times 128 \times 3 = 60 \times 16384 \times 3 = 60 \times 49152 = 2949120$$

Option (A) omits the colour channels ($60 \times 16384 = 983040$). Option (C) is a single RGB frame ($128 \times 128 \times 3 = 49152$). Option (D) is a single grayscale frame ($128 \times 128 = 16384$).

(ii) A model compresses this clip to a 256-dimensional representation using $\mathbf{z} = A\mathbf{x}$. What is the shape of A?

Answer: (B)

Solution. The input is $\mathbf{x} \in \mathbb{R}^{2949120}$ and the output is $\mathbf{z} \in \mathbb{R}^{256}$. Since shape is (output $\times$ input),

$$A \text{ is } 256 \times 2949120$$

Check by matrix multiplication: $(256 \times 2949120)(2949120 \times 1) = (256 \times 1)$. ✓

Option (A) is the transpose — the single most common error. Writing A as (input $\times$ output) makes the multiplication $A\mathbf{x}$ undefined. Option (C) uses a single frame as the input dimension. Option (D) confuses the frame count with a dimension.

(iii) A grayscale image of size 128×128 is flattened into $\mathbf{x}$, and a model computes $\mathbf{y} = A\mathbf{x} + \mathbf{b}$ where $\mathbf{y} \in \mathbb{R}^{64}$. Which is correct?

Answer: (B)

Solution. Grayscale means one channel, so $\mathbf{x} \in \mathbb{R}^{128\times128} = \mathbb{R}^{16384}$. The output is $\mathbf{y} \in \mathbb{R}^{64}$. Therefore:

$$A \text{ is } 64 \times 16384 \qquad \text{and} \qquad \mathbf{b} \in \mathbb{R}^{64}$$

The bias is added to the output, so it must have the output's dimension.

Option (A) transposes A and gives $\mathbf{b}$ the input dimension — both errors at once. Option (C) leaves the image unflattened. Option (D) uses 128 rather than 16384 as the input dimension.

Note. *A reliable check: write the shapes underneath the equation and confirm the inner dimensions cancel.*

$$\underbrace{\mathbf{y}}_{64\times1} = \underbrace{A}_{64\times16384}\underbrace{\mathbf{x}}_{16384\times1} + \underbrace{\mathbf{b}}_{64\times1}$$

The 16384 appears twice adjacently and cancels, leaving 64×1 on both sides.

4 Statistics of a Dataset

Key ideas for this section

For a dataset with n samples and d features, using the **sample** convention:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2, \qquad \mathrm{Cov}(X,Y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$$

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\,\sigma_Y} \in [-1,1], \qquad \mathrm{Var}[X\pm Y] = \mathrm{Var}[X]+\mathrm{Var}[Y]\pm 2\mathrm{Cov}(X,Y), \qquad \mathrm{Var}[aX] = a^2\mathrm{Var}[X]$$

The covariance matrix Σ is $d \times d$ with $\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$. It is symmetric, carries the variances on its diagonal, and is positive semi-definite.

Working for Q1

The dataset is

Sample	X_1	X_2	X_3
1	2	4	1
2	4	6	3
3	6	10	2
4	8	12	6

Means: $\bar{x}_1 = 20/4 = 5, \quad \bar{x}_2 = 32/4 = 8, \quad \bar{x}_3 = 12/4 = 3.$

Centred data (subtract each column's mean):

Sample	$X_1 - \bar{x}_1$	$X_2 - \bar{x}_2$	$X_3 - \bar{x}_3$
1	-3	-4	-2
2	-1	-2	0
3	1	2	-1
4	3	4	3

Dividing every sum of products by $n - 1 = 3$ gives the full covariance matrix, used throughout below:

$$\Sigma = \begin{bmatrix} 6.6667 & 9.3333 & 4.6667 \\ 9.3333 & 13.3333 & 6.0000 \\ 4.6667 & 6.0000 & 4.6667 \end{bmatrix}$$

Q1. A dataset has 4 samples and 3 features (table above). Use the sample convention throughout.

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(i) What is the mean of feature X_2?
Answer: (B)
Solution.
$$\bar{x}_2 = \frac{4+6+10+12}{4} = \frac{32}{4} = 8$$
Option (A) is the mean of the first two values only. Option (C) is the median-adjacent value 10. Option (D) divides by 3 instead of 4 — the $n-1$ correction applies to variance, never to the mean.

(ii) What is the sample variance of feature X_1?
Answer: (B)
Solution. The centred values are $[-3, -1, 1, 3]$, so
$$s_1^2 = \frac{(-3)^2+(-1)^2+1^2+3^2}{4-1} = \frac{9+1+1+9}{3} = \frac{20}{3} \approx 6.6667$$
Option (A) divides by $n=4$ instead of $n-1=3$, giving 5 — this is the *population* variance. Option (C) is the raw sum of squared deviations with no division at all. Option (D) is the standard deviation $\sqrt{20/3} \approx 2.5820$, not the variance.

Note. *Why $n-1$? The deviations are measured about $\bar{x}$, which was itself computed from the same data, so the sum of squared deviations is systematically too small. Dividing by $n-1$ (Bessel's correction) makes the estimator unbiased for the population variance.*

(iii) What is the sample covariance between X_1 and X_2?
Answer: (B)
Solution. Multiply the centred values pairwise and sum:
$$\mathrm{Cov}(X_1, X_2) = \frac{(-3)(-4)+(-1)(-2)+(1)(2)+(3)(4)}{3} = \frac{12+2+2+12}{3} = \frac{28}{3} \approx 9.3333$$
The positive sign says the two features move together: when X_1 is above its mean, so is X_2.

Option (A) divides by 4. Option (C) omits the division entirely. Option (D) is $\mathrm{Var}[X_2]$, i.e. the wrong entry of Σ.

(iv) What is the correlation coefficient between X_1 and X_2?
Answer: (A)
Solution.
$$\rho_{12} = \frac{\mathrm{Cov}(X_1, X_2)}{\sigma_1\sigma_2} = \frac{9.3333}{\sqrt{6.6667}\times\sqrt{13.3333}} = \frac{9.3333}{\sqrt{88.8889}} = \frac{9.3333}{9.4281} \approx 0.9899$$
Very close to $+1$, indicating a nearly perfect positive linear relationship — which the data confirms, since $X_2 \approx 1.4X_1 + 1$ fits all four points closely.

Options (B) and (C) are the other two correlations in the matrix, $\rho_{13} = 0.8367$ and $\rho_{23} = 0.7606$, so they catch students who compute the wrong pair. Option (D) is what you would get if the relationship were exactly linear, which it is not quite.

(v) What are the dimensions of the covariance matrix Σ for this dataset?

Answer: (B)

Solution. Σ is $d \times d$ where d is the number of **features**, so here it is 3×3. The entry $\Sigma_{ij} = \text{Cov}(X_i, X_j)$ has both indices ranging over features, which is why the sample count does not appear.

Option (A) uses $n = 4$, the number of samples — the most common error. Options (C) and (D) mix the two.

Note. *One way to see it: $\Sigma = \frac{1}{n-1} X_c^\top X_c$ where X_c is the centred design matrix of shape $n \times d$. The shapes give $(d \times n)(n \times d) = d \times d$, and the n cancels out entirely.*

(vi) What is $\text{Var}[X_1 + X_2]$?

Answer: (C)

Solution.

$$\text{Var}[X_1 + X_2] = \text{Var}[X_1] + \text{Var}[X_2] + 2\text{Cov}(X_1, X_2)$$

$$= 6.6667 + 13.3333 + 2(9.3333) = 20 + 18.6667 = 38.6667$$

Option (A) is $\text{Var}[X_1] + \text{Var}[X_2] = 20$, forgetting the covariance term entirely — this would be correct only if the features were uncorrelated. Option (B) includes the covariance once instead of twice. Option (D) is $\text{Var}[X_1 - X_2] = 20 - 18.6667 = 1.3333$, using the wrong sign.

Note. *The gap between 38.6667 and 20 is the whole point: positively correlated variables add* more *variance than independent ones, because they tend to be large at the same time. This is the mathematics behind portfolio diversification — combining negatively correlated assets reduces total variance.*

(vii) What is the trace of Σ?

Answer: (A)

Solution. The trace is the sum of the diagonal entries, and the diagonal of a covariance matrix holds the variances:

$$\text{tr}(\Sigma) = \text{Var}[X_1] + \text{Var}[X_2] + \text{Var}[X_3] = 6.6667 + 13.3333 + 4.6667 = 24.6667$$

Option (B) omits $\text{Var}[X_3]$. Options (C) and (D) are individual diagonal entries.

Note. The trace is the total variance *in the dataset. Since the trace also equals the sum of the eigenvalues, and PCA's eigenvalues are the variances along the principal directions, the trace is the quantity that "explained variance ratios" are computed against.*

Q2. Consider a dataset with n samples and d features, with covariance matrix Σ.

(i) Suppose every value of X_1 is multiplied by 100. Which of the following is correct?

Answer: (B)

Solution. Variance. Scaling a random variable scales its variance *quadratically*:

$$\mathrm{Var}[aX] = \mathbb{E}\big[(aX - \mathbb{E}[aX])^2\big] = \mathbb{E}\big[a^2(X - \mathbb{E}[X])^2\big] = a^2\mathrm{Var}[X]$$

With $a = 100$, the variance is multiplied by $100^2 = 10000$.

Correlation. Correlation is scale-invariant. Both the covariance and the standard deviation scale linearly in a:

$$\rho_{aX,Y} = \frac{\mathrm{Cov}(aX, Y)}{\sigma_{aX}\,\sigma_Y} = \frac{a\mathrm{Cov}(X, Y)}{|a|\sigma_X\sigma_Y} = \rho_{X,Y} \quad \text{for } a > 0$$

The factor cancels, so ρ_{12} is unchanged.

Option (A) uses linear rather than quadratic scaling for the variance. Options (C) and (D) wrongly make correlation scale-dependent — but the entire purpose of normalising covariance by the standard deviations is to remove units.

Note. *The quadratic scaling is exactly why PCA on the raw covariance matrix is dominated by whichever feature has the largest units. Measuring salary in rupees rather than lakhs inflates its variance by 10^{10} and the first principal component will point almost entirely along that axis. Standardising the features first — which turns Σ into the correlation matrix — avoids this.*

(ii) Two features X and Y satisfy $\mathrm{Cov}(X, Y) = 0$. What can be concluded?

Answer: (B)

Solution. Zero covariance rules out a *linear* relationship, and nothing more. Correlation measures alignment between two centred vectors, so a curved relationship can be perfectly deterministic while having zero net linear alignment.

Standard counterexample. Let X be uniform on $\{-1, 0, 1\}$ and set $Y = X^2$. Then $\mathbb{E}[X] = 0$ by symmetry, and

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[X^3] - 0 = 0$$

since every odd moment of a symmetric distribution vanishes. Yet Y is a deterministic function of X: knowing X tells you Y exactly. They are maximally dependent with zero covariance.

The implication runs one way only:

$$\text{independent} \implies \text{Cov} = 0, \qquad \text{Cov} = 0 \not\Rightarrow \text{independent}$$

Option (A) reverses this. Options (C) and (D) do not follow at all — two variables can have zero covariance with entirely different distributions and both having large variance.

Note. *There is one important exception. If X and Y are* jointly *Gaussian, then zero covariance does imply independence, because a multivariate normal is fully specified by its mean and covariance, so a zero off-diagonal entry makes the joint density factorise. Note that "jointly Gaussian" is strictly stronger than "each is Gaussian" — the counterexample $Y = SX$ with $S = \pm 1$ has Gaussian marginals, zero covariance, and complete dependence.*

(iii) Given $\text{Var}[X] = 9$, $\text{Var}[Y] = 4$ and $\text{Cov}(X, Y) = 4.5$, what are $\text{Var}[X - Y]$ and $\text{Var}[Y - X]$?

Answer: (C)

Solution.

Var[X - Y] = Var[X] + Var[Y] - 2Cov(X,Y) = 9 + 4 - 2(4.5) = 13 - 9 = 4

Var[Y - X] = Var[Y] + Var[X] - 2Cov(Y,X) = 4 + 9 - 2(4.5) = 4

$$Both equal$$

4.$Twofactsmakethisso: covarianceissymmetric,$Cov(X,Y) = Cov(Y,X)$, andnegationcannotchang$
Z] = (-1)^{2}Var[Z] = Var[Z].

Option (A) suggests the order of subtraction matters. Option (B) is the same error reversed. Option (D) omits the covariance term.

Note. *This is the exact analogue of the norm symmetry $\|\mathbf{v}_1 - \mathbf{v}_2\| = \|\mathbf{v}_2 - \mathbf{v}_1\|$ from the Norms section. In both cases the squaring destroys the sign, so reversing the direction of a difference leaves the magnitude unchanged.*

(iv) Which identity holds for **any** two random variables, with no independence assumption?

Answer: (B)

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Solution. Linearity of expectation, $\mathbb{E}[X+Y]=\mathbb{E}[X]+\mathbb{E}[Y]$, requires no assumptions whatsoever about the joint distribution. This is what makes it such a powerful proof tool — it holds even when X and Y are strongly dependent, or when one is a function of the other.

The other three all acquire a covariance term:

$$\begin{aligned} \mathrm{Var}[X+Y] &= \mathrm{Var}[X]+\mathrm{Var}[Y]+2\mathrm{Cov}(X,Y) && \text{(A) is missing 2Cov} \\ \mathbb{E}[XY] &= \mathbb{E}[X]\mathbb{E}[Y]+\mathrm{Cov}(X,Y) && \text{(C) is missing Cov} \\ \mathrm{Var}[X-Y] &= \mathrm{Var}[X]+\mathrm{Var}[Y]-2\mathrm{Cov}(X,Y) && \text{(D) has the wrong sign on } \mathrm{Var}[Y] \end{aligned}$$

Option (D) is the worst of the three: variances *add* even when the variables are subtracted, so writing a minus sign in front of $\mathrm{Var}[Y]$ is wrong regardless of the covariance.

Note. *Options (A) and (C) are actually equivalent to each other: both hold precisely when* $\mathrm{Cov}(X,Y)=0$. *That condition is weaker than independence, so uncorrelated variables suffice — independence is sufficient but not necessary for either identity.*